

# LUC SARZYNIEC

Senior Software Engineer

## CONTACT

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## SKILLS

### Programming Languages

Java, Python, Perl, Ruby, C, SQL

Personal projects: Crystal, Rust

### Datastore Technologies

DynamoDB, MySQL, ElastiCache,

Cassandra, MongoDB, Elasticsearch

### Streaming Technologies

Kafka, Kinesis, RabbitMQ,

EventBridge, SQS, SNS, Redis

### Other Technologies

Spark, JanusGraph, Gremlin

AWS CDK, Puppet, k8s, Envoy

## EDUCATION

### Master of Research in CS

Machine Learning specialization

Université de Lorraine, 2011

### Master of Computer Engineering

Software Engineering specialization

TELECOM Nancy, 2010

## ONLINE CERTIFICATIONS

Cryptography 1, Stanford University

## PROFESSIONAL SUMMARY

Senior Software Engineer with 15 years of experience in high-scale backend development, system design, and distributed computing. Proven leader in modernizing complex systems, architecting highly extensible and scalable solutions. Proficient across a wide range of programming languages, platforms and technologies.

## WORK EXPERIENCE

### SENIOR SOFTWARE ENGINEER II

Loyalty & Incentives department – Booking.com, Travel industry, Netherlands

2023 - Present

- Modernized systems at the core of Booking.com loyalty program, 'Genius'.
- Leveraged AWS technologies to migrate existing on-prem batch processing systems to event-driven solutions, with extended product capabilities.
- Architected a new highly extensible & scalable progression engine for Genius, enabling better decoupling, new integrations and impactful product features.
- Collaborated with principals and architects to build a target architecture.
- Built & deployed new secured, low-latency services & APIs for high traffic.
- Led large-scale data migrations, including backfilling and data validation.
- Proactively led initiatives to remediate 50+ Security & Compliance gaps.
- Mentored engineers, provided technical guidance, led solutioning sessions.

### SENIOR SOFTWARE ENGINEER

Accommodations Pricing area – Booking.com, Travel industry, Netherlands

2020 - 2022

- Migrated & re-architected the monolithic Perl core of a discounting product to Java services, with shadowing & validation to prevent impact on customers.
- Collaborated to implement new data pipelines, for efficient decision making and enabling machine learning powered dynamic pricing strategies.
- Iterated on dozens of discounting strategy ideas, to match a targeted ROI.
- Created a Bazel mono-repository and CI pipelines to speed up development, improve cross-project/team collaboration, and mitigate dependency issues.
- Led an initiative to improve testing and quality in all the team group's systems.
- Participated in the SOx audit of the product, presenting the tech narrative.

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## PUBLICATIONS

*Design and Evaluation of a Virtual Experimental Environment for Distributed Systems* – PDP'2013, Belfast

*Kadeploy3: Efficient and Scalable Operating System Provisioning for Clusters* – Usenix ;login:, February 2013

*Scalability Testing of the Kadeploy Cluster Deployment System using Virtual Machines on Grid'5000* – CCGRID'2012, Ottawa

*Adding virtualization Capabilities to Grid'5000* – CLOSER'2012, Porto

## PERSONAL PROJECTS

*libgen* – Crystal C-bindings generator

*icu.cr* – Crystal binding for ICU library

*groinc* – Network packet sniffer

*crypto* – Standard crypto. algorithms

...more

## WORK EXPERIENCE

### BACKEND SOFTWARE ENGINEER

Third party inventory area – Booking.com, Travel industry, Netherlands

2018 - 2020

- Piloted the integration of a product into the company's brand new reservation management & payout systems. Built a payout reconciliation engine.
- Implemented reusable CI pipelines to automate software QA & deployment processes, increasing deployments frequency and reducing manual errors.
- Participated in the migration of Perl monolithic logic to Java micro-services.
- Actively shared knowledge on Kubernetes & service management topics.

### SENIOR SOFTWARE ENGINEER

Search engines – Qwant & Xilopix, Tech startups, France

2014 - 2018

- Built a distributed computing framework leveraging RabbitMQ & MongoDB.
- Designed and developed the core of the *Xaphir* search engine: distributed crawlers, scrapers and indexers using the framework and Elasticsearch.
- Created automated product classification algorithms and ontology-based Wikipedia data extraction tools for *Qwant* search engine.
- Built CI pipelines to automate deployments & QA across the company.
- Release manager: weekly release of the engine, including a dozen components.

### SYSTEM SOFTWARE ENGINEER (RESEARCH)

High Performance Computing department – Inria & Loria, CS Research, France

2011 - 2014

- Designed & created '*distem*', a light distributed emulation system (pre-Docker containers deployment, orchestration & configuration for research use-cases).
- Maintained & improved '*kadeploy*', an OS deployment solution for clusters.
- Administration, improvements & development of the *Grid'5000* platform (1000+ machines, 30+ clusters, 10+ sites).